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(54) **FOREIGN MATTER INSPECTION DEVICE AND FOREIGN MATTER INSPECTION METHOD**

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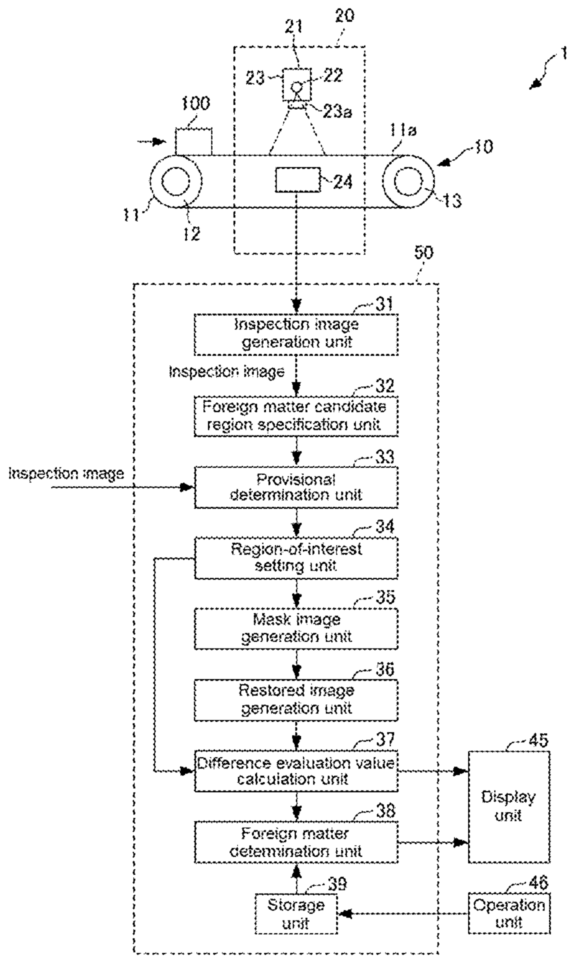
(57) **ABSTRACT**

A foreign matter inspection device includes a region-of-interest setting unit that sets a region of interest including a foreign matter candidate region which is likely to include an image of a foreign matter contained in a product, a mask image generation unit that generates a mask image obtained by masking the foreign matter candidate region, a restored image generation unit that generates a restored image of a good product from the mask image based on a restoration algorithm, a difference evaluation value calculation unit that calculates a difference evaluation value for evaluating a difference between a region-of-interest image and the restored image, and a foreign matter determination unit that determines whether or not the foreign matter candidate region includes the image of the foreign matter based on a comparison between the difference evaluation value and a predetermined threshold value.



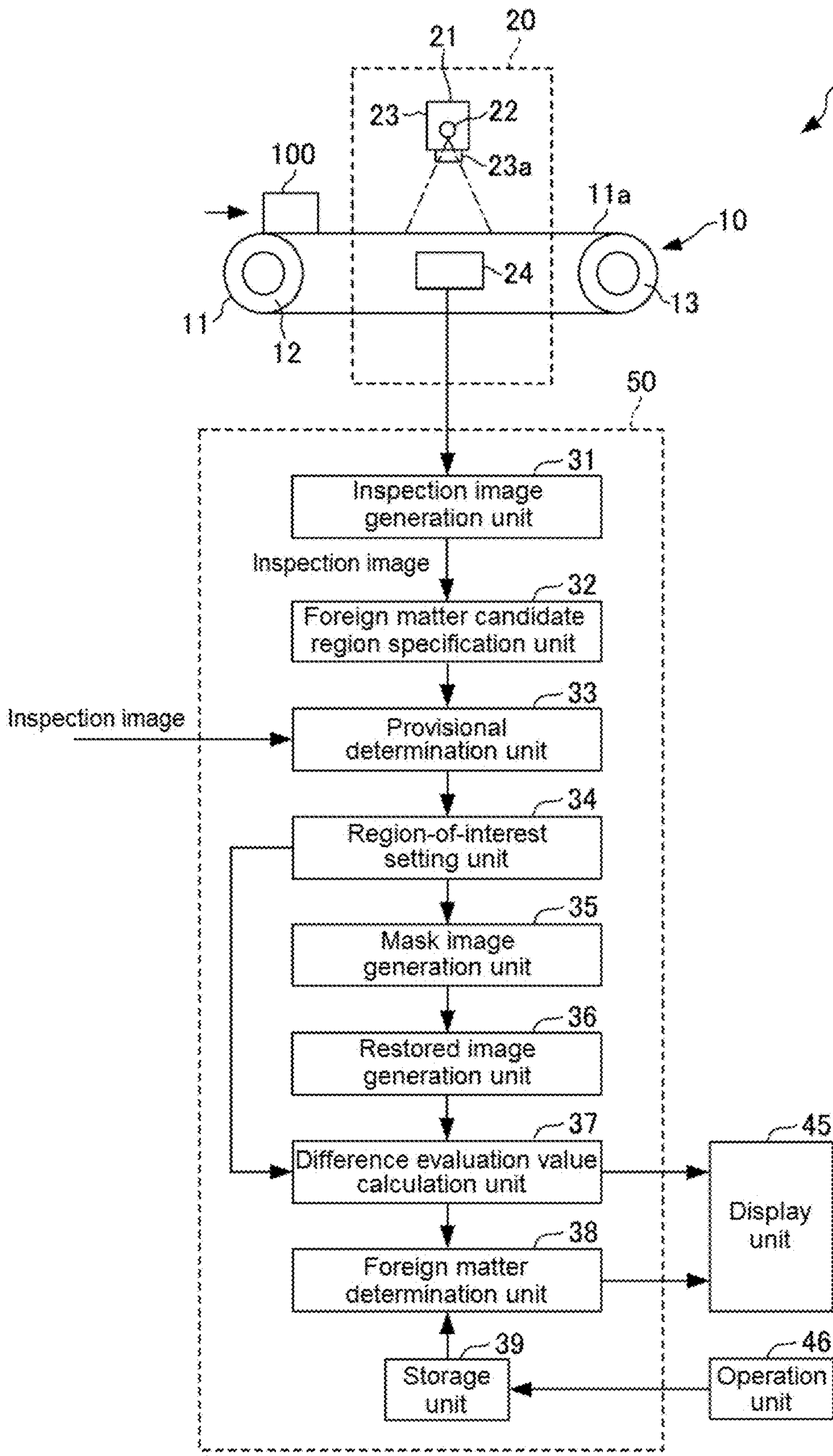

20
21
1
23
22
23a
100
11a
10
24
13
11
12
50
31
Inspection image generation unit
Inspection image
32
Foreign matter candidate region specification unit
33
Inspection image
Provisional determination unit
34
Region-of-interest setting unit
35
Mask image generation unit
36
Restored image generation unit
37
Difference evaluation value calculation unit
45
Display unit
38
Foreign matter determination unit
39
Storage unit
46
Operation unit

FIG. 1

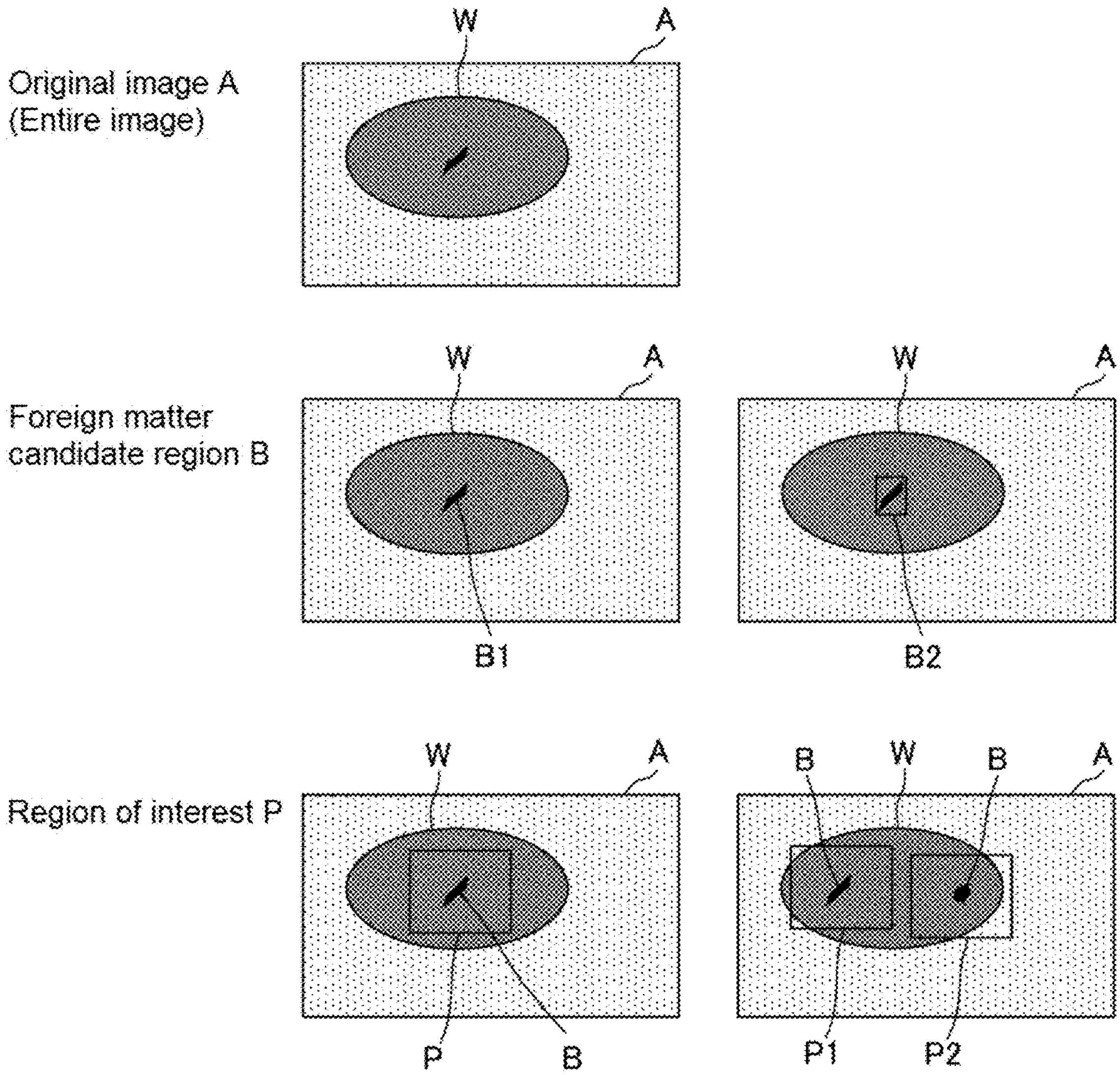

Original image A
(Entire image)
W
A
Foreign matter
candidate region B
W
A
B1
W
A
B2
Region of interest P
W
A
P
B
B
W
B
A
P1
P2

FIG. 2

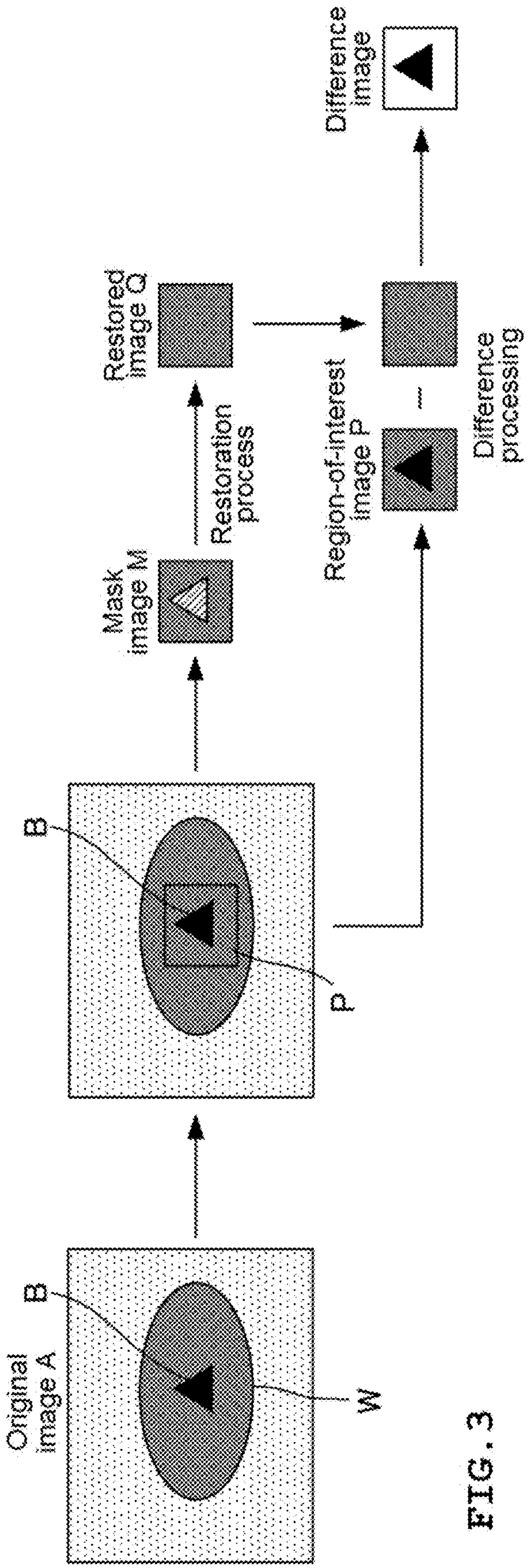

Original image A
B
W
B
P
Mask image M
Restoration process
Restored image Q
Region-of-interest image P
−
Difference processing
Difference image

FIG. 3

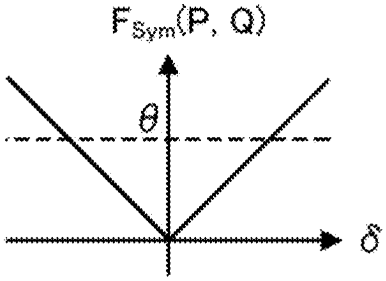


FIG. 4A

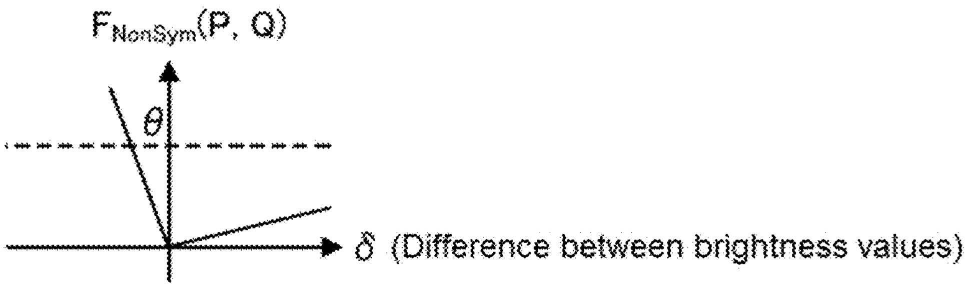


FIG. 4B

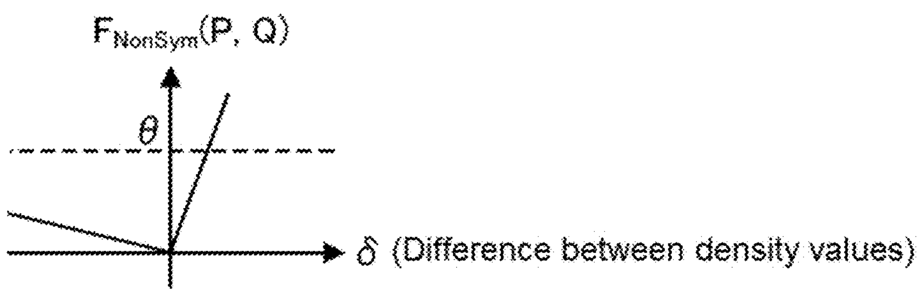


FIG. 4C

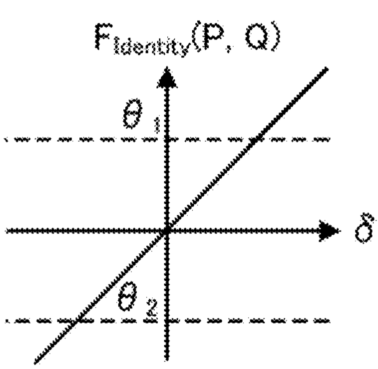


FIG. 4D

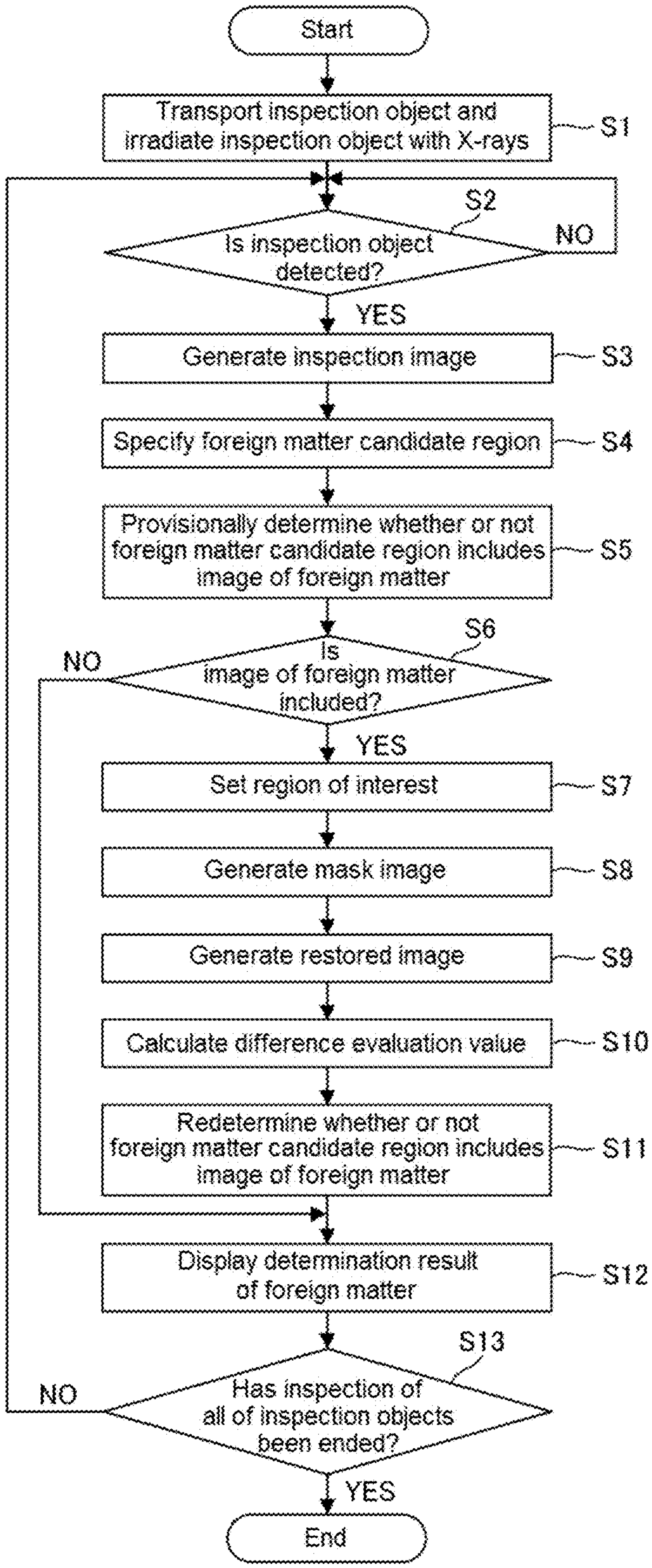

Start
Transport inspection object and irradiate inspection object with X-rays
S1
S2
Is inspection object detected?
NO
YES
Generate inspection image
S3
Specify foreign matter candidate region
S4
Provisionally determine whether or not foreign matter candidate region includes image of foreign matter
S5
S6
Is image of foreign matter included?
NO
YES
Set region of interest
S7
Generate mask image
S8
Generate restored image
S9
Calculate difference evaluation value
S10
Redetermine whether or not foreign matter candidate region includes image of foreign matter
S11
Display determination result of foreign matter
S12
S13
Has inspection of all of inspection objects been ended?
NO
YES
End

FIG. 5

FOREIGN MATTER INSPECTION DEVICE AND FOREIGN MATTER INSPECTION METHOD

TECHNICAL FIELD

[0001] The present invention relates to a foreign matter inspection device and a foreign matter inspection method that inspect a foreign matter in an inspection object.

BACKGROUND ART

[0002] In the related art, as a foreign matter inspection device that detects a foreign matter, such as metal or bone, included in an inspection object, such as food, there is a foreign matter inspection device using X-rays. The X-ray foreign matter inspection device is a device that irradiates an inspection object, such as meat, fish, processed food, or a pharmaceutical product, which is sequentially transported on a transport path at predetermined intervals with X-rays and inspects whether or not a foreign matter is contained in the inspection object from the amount of irradiated X-rays transmitted through the inspection object (for example, see Patent Document 1).

[0003] In the X-ray foreign matter inspection device, a reference value, such as a foreign matter detection reference value or a determination target value, for determining whether or not the foreign matter is contained is set in advance, and a detection value detected by the device is compared with the reference value to determine whether or not the foreign matter is contained. In the case of a determination method using this reference value, there is a concern that erroneous determination will occur depending on the method for setting the reference value, and there is a demand for improvement in the accuracy of determination.

[0004] For example, when the reference value is set to a small value in order to detect the foreign matter with high sensitivity, there is a problem that erroneous detection, in which a good product is determined to be a defective product, increases. Conversely, when the reference value is set to a large value, there is a problem that erroneous detection, in which a defective product is determined to be a good product, increases.

[0005] Therefore, in the device disclosed in Patent Document 1, for a good product that is not determined to have a foreign matter based on a first reference value K_1, a measured value of the good product is compared with a second reference value K_2 set based on the measured value to determine whether or not the foreign matter is likely to be included in the inspection object, and information prompting an inspection operator to reinspect the good product determined to be likely to include the foreign matter is displayed.

RELATED ART DOCUMENT

[Patent Document]

[0006] [Patent Document 1] JP-A-2007-263836

DISCLOSURE OF THE INVENTION

Problem that the Invention is to Solve

[0007] However, the device disclosed in Patent Document 1 only determines the necessity of reinspection and has a problem in that the device does not finally determine whether the inspection object is a good product or a defective product.

[0008] The present invention has been made in order to solve the problems of the related art, and an object of the present invention is to provide a foreign matter inspection device and a foreign matter inspection method that can reduce erroneous detection of a foreign matter and inspect the foreign matter with high accuracy.

Means for Solving the Problem

[0009] In order to solve the above-described problems, according to a first aspect of the present invention, there is provided a foreign matter inspection device that detects a foreign matter included in an inspection object based on an image. The foreign matter inspection device includes: a region-of-interest setting unit that sets a region of interest including a foreign matter candidate region, which is likely to include an image of the foreign matter in the inspection object, in an inspection image in which the foreign matter candidate region has been specified; a mask image generation unit that generates a mask image obtained by masking the foreign matter candidate region in an image of the region of interest; a restored image generation unit that generates a restored image from the mask image based on a restoration algorithm of a model obtained by learning a good product image of the inspection object; a difference evaluation value calculation unit that calculates a difference evaluation value for evaluating a difference between the image of the region of interest and the restored image; and a foreign matter determination unit that determines whether or not the foreign matter candidate region includes the image of the foreign matter based on a comparison between the difference evaluation value and a predetermined threshold value.

[0010] Therefore, the foreign matter inspection device according to the first aspect of the present invention can reduce erroneous detection of the foreign matter and inspect the foreign matter with high accuracy, as compared to a case where the determination of the foreign matter is performed using only the image of the region of interest. For example, when provisional determination is performed using only the image of the region of interest and then redetermination is performed using a difference image between the image of the region of interest and the restored image, it is possible to double-check the inspection and to improve the inspection performance.

[0011] In addition, according to a second aspect of the present invention, in the foreign matter inspection device according to the first aspect, the inspection image may be an X-ray absorption image obtained by converting an amount of X-rays transmitted through the inspection object into a density value proportional to a thickness of the inspection object, and the difference evaluation value may be a value based on a difference between density values of the image of the region of interest and the restored image. When an absolute value of a positive difference between density values is equal to an absolute value of a negative difference between density values, the difference evaluation value obtained from the positive difference between the density values may be different from the difference evaluation value obtained from the negative difference between the density values.

[0012] In the foreign matter inspection device according to the second aspect of the present invention, the difference evaluation value obtained from the positive difference between the density values is set to be larger than the difference evaluation value obtained from the negative difference between the density values having the same absolute value. Therefore, when the inspection image is the X-ray absorption image, it is possible to accurately detect the foreign matter having a positive difference between the density values of the image of the region of interest and the restored image.

[0013] Further, in the foreign matter inspection device according to the second aspect of the present invention, the difference evaluation value obtained from the positive difference between the density values is set to be smaller than the difference evaluation value obtained from the negative difference between the density values having the same absolute value. Therefore, when the inspection image is the X-ray absorption image, it is possible to accurately detect the foreign matter having a negative difference between the density values of the image of the region of interest and the restored image.

[0014] In addition, according to a third aspect of the present invention, in the foreign matter inspection device according to the first aspect, the inspection image may be an X-ray transmission image obtained by converting an amount of X-rays transmitted through the inspection object into a brightness value, and the difference evaluation value is a value based on a difference between brightness values of the image of the region of interest and the restored image. When an absolute value of a positive difference between brightness values is equal to an absolute value of a negative difference between brightness values, the difference evaluation value obtained from the negative difference between the brightness values may be different from the difference evaluation value obtained from the positive difference between the brightness values.

[0015] In the foreign matter inspection device according to the third aspect of the present invention, the difference evaluation value obtained from the negative difference between the brightness values is set to be larger than the difference evaluation value obtained from the positive difference between the brightness values having the same absolute value. Therefore, when the inspection image is the X-ray transmission image, it is possible to accurately detect the foreign matter having a negative difference between the brightness values of the image of the region of interest and the restored image.

[0016] In addition, in the foreign matter inspection device according to the third aspect of the present invention, the difference evaluation value obtained from the negative difference between the brightness values is set to be smaller than the difference evaluation value obtained from the positive difference between the brightness values having the same absolute value. Therefore, when the inspection image is the X-ray transmission image, it is possible to accurately detect the foreign matter having a positive difference between the brightness values of the image of the region of interest and the restored image.

[0017] Further, according to a fourth aspect of the present invention, the foreign matter inspection device according to the first aspect may further include a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image. Furthermore, according to a fifth aspect of the present invention, the foreign matter inspection device according to the second aspect may further include a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image. Moreover, according to a sixth aspect of the present invention, the foreign matter inspection device according to the third aspect may further include a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image.

[0018] In addition, according to a seventh aspect of the present invention, there is provided a foreign matter inspection method for detecting a foreign matter included in an inspection object based on an image. The foreign matter inspection method includes: a region-of-interest setting step of setting a region of interest including a foreign matter candidate region, which is likely to include an image of the foreign matter in the inspection object, in an inspection image in which the foreign matter candidate region has been specified; a mask image generation step of generating a mask image obtained by masking the foreign matter candidate region in an image of the region of interest; a restored image generation step of generating a restored image from the mask image based on a restoration algorithm of a model obtained by learning a good product image of the inspection object; a difference evaluation value calculation step of calculating a difference evaluation value for evaluating a difference between the image of the region of interest and the restored image; and a foreign matter determination step of determining whether or not the foreign matter candidate region includes the image of the foreign matter based on a comparison between the difference evaluation value and a predetermined threshold value.

Advantage of the Invention

[0019] Therefore, the foreign matter inspection method according to the seventh aspect of the present invention can reduce erroneous detection of the foreign matter and inspect the foreign matter with high accuracy, as compared to a case where the determination of the foreign matter is performed using only the image of the region of interest.

[0020] In addition, according to an eighth aspect of the present invention, in the foreign matter inspection method according to the seventh aspect, the inspection image may be an X-ray absorption image obtained by converting an amount of X-rays transmitted through the inspection object into a density value proportional to a thickness of the inspection object, and the difference evaluation value may be a value based on a difference between density values of the image of the region of interest and the restored image. When an absolute value of a positive difference between density values is equal to an absolute value of a negative difference between density values, the difference evaluation value obtained from the positive difference between the density values may be different from the difference evaluation value obtained from the negative difference between the density values.

[0021] Further, according to a ninth aspect of the present invention, in the foreign matter inspection method according to the seventh aspect, the inspection image may be an X-ray transmission image obtained by converting an amount of X-rays transmitted through the inspection object into a brightness value, and the difference evaluation value may be a value based on a difference between brightness values of the image of the region of interest and the restored image. When an absolute value of a positive difference between brightness values is equal to an absolute value of a negative difference between brightness values, the difference evaluation value obtained from the negative difference between the brightness values may be different from the difference evaluation value obtained from the positive difference between the brightness values.

[0022] Furthermore, according to a tenth aspect of the present invention, the foreign matter inspection method according to the seventh aspect may further include a foreign matter candidate region specification step of specifying the foreign matter candidate region in the inspection image.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. **1** is a schematic diagram showing a configuration of a foreign matter inspection device according to an embodiment of the present invention.

[0024] FIG. **2** is a diagram showing a region of an inspection image handled by the foreign matter inspection device shown in FIG. **1**.

[0025] FIG. **3** is a diagram showing a process of the foreign matter inspection device shown in FIG. **1**.

[0026] FIGS. **4**A to **4**D are graphs showing a relationship between a difference evaluation value calculated by a difference evaluation value calculation unit provided in the foreign matter inspection device shown in FIG. **1** and a difference between brightness values or density values of a region-of-interest image and a restored image, FIG. **4**A shows an example in which the difference evaluation value is symmetrical with respect to the positive and negative values of the difference between the brightness values or the density values, FIGS. **4**B and **4**C show an example in which the difference evaluation value is asymmetrical with respect to the positive and negative values of the difference between the brightness values or the density values, and FIG. **4**D shows an example in which the difference evaluation value is equal to the difference between the brightness values or the density values.

[0027] FIG. **5** is a flowchart showing a process of a foreign matter inspection method using the foreign matter inspection device shown in FIG. **1**.

BEST MODE FOR CARRYING OUT THE INVENTION

[0028] Hereinafter, an embodiment of a foreign matter inspection device and a foreign matter inspection method according to the present invention will be described with reference to the drawings.

[0029] A foreign matter inspection device **1** is installed in a portion of a transport line for transporting an inspection object **100** and detects the presence or absence of a foreign matter, such as metal, glass, stone, or bone, contained in the inspection object **100** that is sequentially transported at a predetermined interval. Examples of the foreign matter inspection device **1** include an X-ray inspection device using X-rays and an inspection device that irradiates the inspection object **100** with light in a near-infrared region or a visible light region and measures transmitted light or reflected light. In the present embodiment, a case where the foreign matter inspection device **1** is the X-ray inspection device will be described as an example.

[0030] As shown in FIG. **1**, the foreign matter inspection device **1** includes a transport unit **10**, an X-ray inspection unit **20**, and a control unit **50** including a display unit **45** and an operation unit **46**.

[0031] The transport unit **10** is a conveyor in which a loop-shaped transport belt **11** is wound around a plurality of transport rollers **12** and **13** and is supported by a housing (not shown). The transport unit **10** can sequentially transport the inspection object **100** placed on a transport surface **11***a* of the transport belt **11** in the right direction in FIG. **1** such that the inspection object **100** passes through a predetermined inspection section of the X-ray inspection unit **20**. The placement of the inspection object **100** on the transport belt **11** may be manually performed by a user or may be performed by a dedicated device (not shown).

[0032] The transport belt **11** is made of a material (an element other than an element with a large atomic mass) that easily transmits X-rays. The transport belt **11** is driven by controlling the rotation of a driving motor such that the transport belt **11** is moved at a transport speed set in advance by the operation unit **46** when the inspection object **100** is inspected. As a result, the inspection object **100** carried in from a carry-in port is transported to a carry-out port in the right direction of FIG. **1** at the set transport speed.

[0033] The X-ray inspection unit **20** includes an X-ray source **21** that irradiates the inspection object **100** transported by the transport unit **10** with the X-rays in a predetermined energy band transmitted through the inspection object **100**. The X-ray source **21** can generate the X-rays with a wavelength and intensity corresponding to a tube current and a tube voltage of a known X-ray tube **22** using the X-ray tube **22** and irradiate the inspection object **100** on the transport belt **11** with fan-beam X-rays in a direction orthogonal to a transport direction of the transport unit **10** through an X-ray window portion **23***a* of an envelope **23**.

[0034] The X-ray inspection unit **20** further includes an X-ray detector **24** that is disposed directly below the transport belt **11**.

[0035] The X-ray detector **24** is configured by an X-ray line sensor camera in which detection elements, each of which includes a scintillator as a phosphor and a photodiode or a charge-coupled device, are disposed in an array at a predetermined pitch in a width direction of a transport path of the transport unit **10** to detect the X-rays at a predetermined resolution, which is not shown.

[0036] That is, the X-ray detector **24** can detect the X-rays, which have been emitted from the X-ray source **21** and transmitted through the inspection object **100**, for each predetermined transmission region of the inspection object **100** corresponding to the detection element, convert the detected X-rays into an electric signal corresponding to a transmission amount of the detected X-rays, and output an X-ray detection signal for each transmission region.

[0037] The control unit **50** controls the transport speed, transport interval, and the like of the inspection object **100** by the transport belt **11** in the transport unit **10**. Further, the control unit **50** controls the X-ray irradiation intensity and irradiation period of the X-ray inspection unit **20** or controls the X-ray detection period of the X-ray line sensor of the X-ray detector **24**, the detection period of the inspection object **100**, and the like according to the transport speed of the inspection object **100**.

[0038] In addition, the control unit **50** includes an inspection image generation unit **31**, a foreign matter candidate region specification unit **32**, a provisional determination unit **33**, a region-of-interest setting unit **34**, a mask image generation unit **35**, a restored image generation unit **36**, a difference evaluation value calculation unit **37**, a foreign matter determination unit **38**, and a storage unit **39**.

[0039] The inspection image generation unit **31** receives the X-ray detection signal from the X-ray detector **24** at each predetermined period and generates an inspection image of the inspection object **100** which consists of information of a two-dimensional position determined by the passing direction of the inspection object **100** and the arrangement direction of the detection elements and a result of signal processing at each position.

[0040] In addition, the inspection image generated by the inspection image generation unit **31** may be an X-ray transmission image obtained by converting the amount of X-rays transmitted through the inspection object **100** into a brightness value or may be an X-ray absorption image obtained by converting the amount of X-rays transmitted through the inspection object **100** into a density value proportional to the thickness of the inspection object **100**. The inspection image generation unit **31** generates the inspection image each time the inspection object **100** is transported to the X-ray inspection unit **20** and passes through a predetermined inspection section.

[0041] For the X-ray absorption image, for example, when an X-ray absorption rate of the inspection object **100** is a and the thickness of the inspection object **100** is L, intensity S' after the X-ray with intensity S is transmitted through the inspection object **100** can be theoretically written as $S'=S\cdot\exp(-\alpha\cdot L)$. When the logarithm of both sides is taken and the expression is transformed, the expression can also be written as $\alpha\cdot L=\log(S)-\log(S')$. The X-ray transmission image corresponds to a two-dimensional distribution of S'. When the logarithm is taken and the expression is transformed as described above, it is possible to calculate the X-ray absorption image showing the two-dimensional distribution of the absorption amount a L by the inspection object **100**.

[0042] Alternatively, the inspection image generated by the inspection image generation unit **31** may be a transmission image or a reflection image of the inspection object **100** captured by the imaging device in a state in which the inspection object **100** is irradiated with light in the near-infrared region or the visible light region.

[0043] FIG. **2** is a diagram showing each region of the inspection image including an image W of the inspection object **100**.

[0044] An original image A is an image including the image W of the inspection object **100** among the inspection images generated by the inspection image generation unit **31**. Further, hereinafter, the original image A of the inspection image is also simply referred to as an "inspection image A".

[0045] A foreign matter candidate region B is defined as a region B**1** that is likely to include an image of a foreign matter in the inspection object **100**, a rectangular region B**2** including the region B**1**, or the like in the original image A.

[0046] A region of interest P is a region including the foreign matter candidate region B in the original image A. When there are a plurality of foreign matter candidate regions B in the original image A, the region of interest P is set for each of the plurality of foreign matter candidate regions B. FIG. **2** shows an example in which one region of interest P is set for one foreign matter candidate region B and an example in which two regions of interest P**1** and P**2** are set for two foreign matter candidate regions B, respectively. In addition, the region of interest P may be the entire region of the original image A. Further, hereinafter, the image of the region of interest P is also referred to as a "region-of-interest image P".

[0047] The foreign matter candidate region specification unit **32** specifies the foreign matter candidate region B that is likely to include the image of the foreign matter in the inspection object **100** in the inspection image A generated by the inspection image generation unit **31**. The specification of the foreign matter candidate region B by the foreign matter candidate region specification unit **32** is performed by various known foreign matter inspection algorithms.

[0048] For example, the foreign matter candidate region specification unit **32** may execute image processing, such as filtering for extracting the image of the foreign matter, on the inspection image A generated by the inspection image generation unit **31** to specify the region of the image of the foreign matter contained in the inspection object **100**. For example, a feature extraction filter, such as a differential filter (a Roberts filter, a Prewitt filter, or a Sobel filter) or a Laplacian filter, is used as a filter for enhancing the image of the foreign matter.

[0049] Alternatively, the foreign matter candidate region specification unit **32** may specify the foreign matter candidate region B in the inspection image A using a deep learning-based object detection model such as a single shot multibox detector (SSD) or you only look once (YOLO).

[0050] The provisional determination unit **33** compares the brightness value or density value of the foreign matter candidate region B specified by the foreign matter candidate region specification unit **32** with a predetermined threshold value to provisionally determine whether or not the image of the foreign matter is included in the foreign matter candidate region B. In addition, the inspection image A provisionally determined to include the foreign matter candidate region B by the provisional determination unit **33** may be an image in which the foreign matter candidate region B has been specified by any external device.

[0051] As schematically shown in FIG. **3**, the region-of-interest setting unit **34** sets the region of interest P including the foreign matter candidate region B in the original image A of the inspection image in which the foreign matter candidate region B has been specified. The setting of the region of interest P by the region-of-interest setting unit **34** may be performed on all of the inspection images A or may be performed only on the inspection image A provisionally determined to include the image of the foreign matter by the provisional determination unit **33**. In addition, when the foreign matter candidate region specification unit **32** specifies a plurality of foreign matter candidate regions B in one inspection image A, the region-of-interest setting unit **34** sets a plurality of regions of interest P respectively including the plurality of foreign matter candidate regions B in the inspection image A.

[0052] As schematically shown in FIG. **3**, the mask image generation unit **35** generates a mask image M subjected to masking the foreign matter candidate region B in the region-of-interest image P set by the region-of-interest setting unit **34**. For example, the mask image generation unit **35** generates, as the mask image M, an image obtained by filling the foreign matter candidate region B in the region-of-interest image P with a constant value of, typically, 0 or by removing the foreign matter candidate region B.

[0053] The restored image generation unit **36** performs a restoration process of generating a restored image Q from the mask image M based on a restoration algorithm of a model obtained by learning a good product image, which is an image of a good product of the inspection object **100**, as schematically shown in FIG. **3**. Here, the good product image is, for example, an image of a good product that is the same type as the inspection object **100** and does not include the foreign matter.

[0054] The restoration algorithm used by the restored image generation unit **36** to generate the restored image Q is, for example, an algorithm using deep learning or an algorithm using K-singular value decomposition (K-SVD).

[0055] The learning of the restored image generation unit **36** is performed using a plurality of good product images subjected to any masking. In this case, it is desirable that the good product image does not include image information, such as a background of the good product, other than the good product.

[0056] That is, the restoration algorithm of the restored image generation unit **36** can learn the good product image that does not include the foreign matter to convert the mask image M into the restored image Q that does not include the foreign matter.

[0057] The difference evaluation value calculation unit **37** calculates a difference evaluation value for evaluating a difference between the region-of-interest image P and the restored image Q generated by the restored image generation unit **36**.

[0058] For example, the difference evaluation value calculation unit **37** calculates, as the difference evaluation value, an absolute value $F_{Sym}(P, Q)$ of a difference δ between the brightness values or density values of the region-of-interest image P and the restored image Q according to the following Expressions (1a), (1b), and (2). FIG. **4**A is a graph showing a difference evaluation value $F_{Sym}(P, Q)$ that is symmetric with respect to the positive and negative values of the difference δ between the brightness values or density values of the region-of-interest image P and the restored image Q.

[Equation 1]

$$i^*, j^* = \mathrm{argmax}_{i,j}|P_{i,j} - Q_{i,j}| \tag{1a}$$

$$\delta = P_{i^*,j^*} - Q_{i^*,j^*} \tag{1b}$$

[Equation 2]

$$F_{Sym}(P, Q) = |\delta| \tag{2}$$

[0059] Expression (1a) indicates pixels (i*, j*) of which the absolute value of the difference δ between the brightness values or density values is the maximum value among the pixels at the same positions in the region-of-interest image P and the restored image Q. δ of Expression (1b) indicates the difference between the brightness values or density values of the region-of-interest image P and the restored image Q in the pixels (i*, j*).

[0060] That is, as schematically shown in FIG. **3**, the difference evaluation value calculation unit **37** performs difference processing of generating a difference image that consists of the difference δ between the brightness values or density values of the region-of-interest image P and the restored image Q. The difference image generated by the difference evaluation value calculation unit **37** may be displayed on the display unit **45**.

[0061] Alternatively, the difference evaluation value calculation unit **37** may calculate, as the difference evaluation value, $F_{NonSym}(P, Q)$ that is asymmetric with respect to the positive and negative values of the difference δ between the brightness values or density values of the region-of-interest image P and the restored image Q according to the following Expression (3).

[Equation 3]

$$F_{NonSym}(P, Q) = w_1\delta + w_2|\delta| \tag{3}$$

[0062] FIG. **4**B is a graph showing a difference evaluation value $F_{NonSym}(P, Q)$ when δ is the difference between the brightness values of the region-of-interest image P and the restored image Q in the pixels (i*, j*). Here, $w_1+w_2>0$, $w_1<w_2$, and $w_1<0$ are satisfied. That is, the absolute value of the slope of the difference evaluation value $F_{NonSym}(P, Q)$ with respect to the negative difference δ between the brightness values is larger than the absolute value of the slope of the difference evaluation value $F_{NonSym}(P, Q)$ with respect to the positive difference δ between the brightness values.

[0063] That is, for $F_{NonSym}(P, Q)$ satisfying $w_1+w_2>0$, $w_1<w_2$, and $w_1<0$, when the absolute value of the positive difference δ between the brightness values is equal to the absolute value of the negative difference δ between the brightness values, the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the negative difference δ between the brightness values is larger than the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the positive difference δ between the brightness values. In the X-ray transmission image, the brightness value of the foreign matter is often smaller than that of other normal portions. Therefore, when $F_{NonSym}(P, Q)$ is used as the difference evaluation value shown in FIG. **4**B, it is possible to improve the accuracy of determining the foreign matter in the foreign matter determination unit **38** in the subsequent stage.

[0064] FIG. **4**C is a graph showing the difference evaluation value $F_{NonSym}(P, Q)$ when δ is the difference between the density values of the region-of-interest image P and the restored image Q in the pixels (i*, j*). Here, $w_1+w_2>0$, $w_1<w_2$, and $w_1>0$ are satisfied. That is, the absolute value of the slope of the difference evaluation value $F_{NonSym}(P, Q)$ with respect to the positive difference δ between the density values is larger than the absolute value of the slope of the difference evaluation value $F_{NonSym}(P, Q)$ with respect to the negative difference δ between the density values.

[0065] That is, for $F_{NonSym}(P, Q)$ in which $w_1+w_2>0$, $w_1<w_2$, and $w_1>0$ are satisfied, when the absolute value of the positive difference δ between the density values is equal to the absolute value of the negative difference δ between the density values, the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the positive difference δ between the density values is larger than the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the negative difference δ between the density values. In the X-ray absorption image, the density value of the foreign matter is often larger than that of other normal portions. Therefore, when $F_{NonSym}(P, Q)$ is used as the difference evaluation value shown in FIG. **4**C, it is possible to improve the accuracy of determining the foreign matter in the foreign matter determination unit **38** in the subsequent stage.

[0066] As described above, the graphs of FIGS. **4**B and **4**C show the difference evaluation values $F_{NonSym}(P, Q)$ suitable for a case where the brightness value of the foreign matter is smaller than that of other normal portions in the X-ray transmission image and a case where the density value of the foreign matter is larger than that of other normal portions in the X-ray absorption image, respectively.

[0067] However, for example, when the user wants detect a cavity in the inspection object **100** as the foreign matter, conversely, the brightness value of the cavity is larger than that of other normal portions in the X-ray transmission image, and the density value of the cavity is smaller than that of other normal portions in the X-ray absorption image.

[0068] In this case, for the X-ray transmission image, when the absolute value of the positive difference δ between the brightness values is equal to the absolute value of the negative difference δ between the brightness values, the difference evaluation value $F_{NonSym}(P, Q)$ at which the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the negative difference δ between the brightness values is smaller than the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the positive difference δ between the brightness values may be used. Similarly, for the X-ray absorption image, when the absolute value of the positive difference δ between the density values is equal to the absolute value of the negative difference δ between the density values, the difference evaluation value $F_{NonSym}(P, Q)$ at which the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the positive difference δ between the density values is smaller than the difference evaluation value $F_{NonSym}(P, Q)$ obtained from the negative difference δ between the density values and may be used.

[0069] Alternatively, the difference evaluation value calculation unit **37** may output the difference δ between the brightness values or density values of the region-of-interest image P and the restored image Q as a difference evaluation value $F_{Identity}(P, Q)$ without any change according to the following Expression (4). FIG. **4**D is a graph showing the difference evaluation value $F_{Identity}(P, Q)$.

[Equation 4]

$$F_{Identity}(P, Q) = \delta \tag{4}$$

[0070] Alternatively, the difference evaluation value calculation unit **37** may calculate a difference evaluation value δ_{SSIM} using a structural similarity index measure (SSIM) according to the following Expressions (5a) to (5d).

[Equation 5]

$$\delta_{SSIM} = SSIM(P, Q) = I(P, Q)c(P, Q)s(P, Q) = \frac{(2\mu_P\mu_Q + c_1)(2\sigma_{PQ} + c_2)}{(\mu_P^2 + \mu_Q^2 + c_1)(\sigma_P^2 + \sigma_Q^2 + c_2)} \tag{5a}$$

$$I(P, Q) = \frac{2\mu_P\mu_Q + c_1}{\mu_P^2 + \mu_Q^2 + c_1} \tag{5b}$$

-continued

$$c(P, Q) = \frac{2\sigma_P\sigma_Q + c_2}{\sigma_P^2 + \sigma_Q^2 + c_2} \tag{5c}$$

$$s(P, Q) = \frac{2\sigma_{PQ} + c_2}{2\sigma_P\sigma_Q + c_2} \tag{5d}$$

[0071] In Expressions (5a) to (5d), μ_P is defined as an average brightness value or average density value of the region-of-interest image P, and μ_Q is defined as an average brightness value or average density value of the restored image Q. σ_P is defined as a standard deviation of the brightness values or density values of the region-of-interest image P, and σ_Q is defined as a standard deviation of the brightness values or density values of the restored image Q. c_1 and c_2 are defined as small constants for preventing division by zero error.

[0072] I(P, Q) indicates a rate of match between the average brightness values or average density values of the region-of-interest image P and the restored image Q and has a minimum value of 0 and a maximum value of 1. c(P, Q) indicates a rate of match between contrasts of the brightness values or density values of the region-of-interest image P and the restored image Q and has a minimum value of 0 and a maximum value of 1. s(P, Q) indicates a correlation coefficient between the brightness values or density values of the region-of-interest image P and the restored image Q and has a minimum value of −1 and a maximum value of 1. That is, the minimum value of the difference evaluation value δ_{SSIM} is −1, and the maximum value thereof is 1.

[0073] Alternatively, the difference evaluation value calculation unit **37** may calculate a similarity (distance) between distributions of the brightness values or density values of the region-of-interest image P and the restored image Q as the difference evaluation value like Kullback-Leibler divergence (KLD). Alternatively, the difference evaluation value calculation unit **37** may calculate a distance between any image feature amounts of the region-of-interest image P and the restored image Q as the difference evaluation value.

[0074] The foreign matter determination unit **38** determines whether or not the foreign matter candidate region B includes the image of the foreign matter based on the comparison between the difference evaluation value calculated by the difference evaluation value calculation unit **37** and a predetermined threshold value. A determination result of the foreign matter determination unit **38** is displayed on the display unit **45**.

[0075] For example, when the difference evaluation value $F_{Sym}(P, Q)$ or $F_{NonSym}(P, Q)$ exceeds a predetermined threshold value θ, the foreign matter determination unit **38** determines that the foreign matter candidate region B includes the image of the foreign matter.

[0076] In addition, in a case where the inspection image A is an X-ray absorption image and the density value of the foreign matter desired to be detected is expected to be larger than that of other normal portions, when $F_{Identity}(P, Q)$ is larger than a predetermined threshold value θ_1 (>0), the foreign matter determination unit **38** determines that the foreign matter candidate region B includes the image of the foreign matter. Meanwhile, in a case where the inspection image A is an X-ray absorption image and the density value of the foreign matter desired to be detected is expected to be smaller than that of other normal portions, when $F_{Identity}$(P,

Q) is smaller than a predetermined threshold value θ_2 (<0), the foreign matter determination unit **38** determines that the foreign matter candidate region B includes the image of the foreign matter.

[0077] In addition, in a case where the inspection image A is an X-ray transmission image and the brightness value of the foreign matter desired to be detected is expected to be larger than that of other normal portions, when $F_{Identity}$(P, Q) is larger than the predetermined threshold value θ_1 (>0), the foreign matter determination unit **38** determines that the foreign matter candidate region B includes the image of the foreign matter. Meanwhile, in a case where the inspection image A is an X-ray transmission image and the brightness value of the foreign matter desired to be detected is expected to be smaller than that of other normal portions, when $F_{Identity}$(P, Q) is smaller than the predetermined threshold value θ_2 (<0), the foreign matter determination unit **38** determines that the foreign matter candidate region B includes the image of the foreign matter.

[0078] FIGS. **4**A to **4**D show a case where the threshold values θ, θ_1, and θ_2 are constant values. However, these threshold values may change depending on the magnitude of the difference δ between the brightness values or the density values.

[0079] Alternatively, the foreign matter determination unit **38** may provide a threshold value for each of various difference evaluation values, such as the difference evaluation value δ_{SSIM} by SSIM and the difference evaluation value by KLD, calculated by the difference evaluation value calculation unit **37** and may determine whether or not the foreign matter candidate region B includes the image of the foreign matter. In addition, the foreign matter determination unit **38** may determine that the foreign matter candidate region B includes the image of the foreign matter only when all of the plurality of difference evaluation values calculated by the difference evaluation value calculation unit **37** are values indicating that the foreign matter candidate region B includes the image of the foreign matter.

[0080] The storage unit **39** stores each threshold value used in the foreign matter determination unit **38**. For example, each threshold value used in the foreign matter determination unit **38** is predetermined according to the type of the inspection object **100** and the type of the difference evaluation value calculated by the difference evaluation value calculation unit **37**. In addition, the threshold value stored in the storage unit **39** may be changed to a desired value by the input of an operation to the operation unit **46** by the user.

[0081] Alternatively, each threshold value used in the foreign matter determination unit **38** may be automatically set based on the inspection image A. For example, the foreign matter determination unit **38** may set each threshold value based on the maximum and minimum values of the brightness value or density value of the inspection image A.

[0082] Hereinafter, an example of a process of the foreign matter inspection method using the foreign matter inspection device **1** according to the present embodiment will be described with reference to a flowchart shown in FIG. **5**. In addition, a description that overlaps with the description of the configuration of the foreign matter inspection device **1** will be omitted as appropriate.

[0083] First, when the user inputs an operation of issuing a measurement start instruction using the operation unit **46**, the transport unit **10** starts sequentially transporting one or more inspection objects **100** placed on the transport belt **11** by the user or the dedicated device. Then, the X-ray source **21** irradiates the inspection object **100**, which is transported by the transport unit **10** and passes through a predetermined inspection section, with the X-rays (Step S**1**).

[0084] Then, when the detection sensor (not shown) detects the entry of the inspection object **100** into the inspection section (Step S**2**: YES), the inspection image generation unit **31** generates an inspection image of the inspection object **100** transported by the transport unit **10** (Step S**3**).

[0085] Then, the foreign matter candidate region specification unit **32** specifies the foreign matter candidate region B that is likely to include the image of the foreign matter in the inspection object **100** in the inspection image A generated by the inspection image generation unit **31** (Step S**4**).

[0086] Then, the provisional determination unit **33** compares the brightness value or density value of the foreign matter candidate region B specified by the foreign matter candidate region specification unit **32** with a predetermined threshold value to provisionally determines whether or not the image of the foreign matter is included in the foreign matter candidate region B (Step S**5**).

[0087] When the provisional determination unit **33** provisionally determines that the image of the foreign matter is included in the foreign matter candidate region B (Step S**6**: YES), a process in Step S**7** and the subsequent steps is executed. When the provisional determination unit **33** provisionally determines that the image of the foreign matter is not included in the foreign matter candidate region B, a process in Step S**12** and the subsequent steps is executed.

[0088] In Step S**7**, the region-of-interest setting unit **34** sets the region of interest P including the foreign matter candidate region B in the inspection image A in which the foreign matter candidate region B has been specified (region-of-interest setting step S**7**).

[0089] Then, the mask image generation unit **35** generates the mask image M obtained by masking the foreign matter candidate region B in the region-of-interest image P (mask image generation step S**8**).

[0090] Then, the restored image generation unit **36** generates the restored image Q from the mask image M based on the restoration algorithm of the model obtained by learning the good product image of the inspection object **100** (restored image generation step S**9**).

[0091] Then, the difference evaluation value calculation unit **37** calculates the difference evaluation value for evaluating the difference between the region-of-interest image P and the restored image Q (difference evaluation value calculation step S**10**).

[0092] Then, the foreign matter determination unit **38** redetermines whether or not the foreign matter candidate region B includes the image of the foreign matter based on the comparison between the difference evaluation value and the predetermined threshold value (foreign matter determination step S**11**).

[0093] Then, the display unit **45** displays the determination result by the provisional determination unit **33** or the foreign matter determination unit **38** (Step S**12**). That is, when the provisional determination unit **33** provisionally determines that the image of the foreign matter is not included in the foreign matter candidate region B, the display unit **45** displays information indicating that the foreign matter is not included in the inspection object **100**.

On the other hand, when the provisional determination unit **33** and the foreign matter determination unit **38** determine that the image of the foreign matter is included in the foreign matter candidate region B, the display unit **45** displays information indicating that the foreign matter is included in the inspection object **100**.

[0094] Then, when the determination results are acquired for all of the inspection objects **100**, that is, when the inspection of all of the inspection objects **100** is ended (Step S**13**: YES), the series of processes is ended. When the inspection of all of the inspection objects **100** is not ended (Step S**13**: NO), the process in Step S**2** and the subsequent steps is executed again.

[0095] As described above, the foreign matter inspection device **1** according to the present embodiment restores the mask image M obtained by masking the foreign matter candidate region B in the region-of-interest image P based on the restoration algorithm and performs the foreign matter determination based on the difference between the obtained restored image Q and the original region-of-interest image P.

[0096] Therefore, the foreign matter inspection device **1** according to the present embodiment can reduce the erroneous detection of the foreign matter and inspect the foreign matter with high accuracy, as compared to a case where the foreign matter determination is performed using only the region-of-interest image P.

[0097] For example, when the foreign matter inspection device **1** according to the present embodiment performs the provisional determination using only the region-of-interest image P and then performs the redetermination using the difference image between the region-of-interest image P and the restored image Q, it is possible to double-check the inspection and to improve the inspection performance.

[0098] In addition, in the foreign matter inspection device **1** according to the present embodiment, the difference evaluation value obtained from the positive difference δ between the density values is set to be larger than the difference evaluation value obtained from the negative difference δ between the density values having the same absolute value. Therefore, when the inspection image A is the X-ray absorption image, it is possible to accurately detect the foreign matter having a positive difference between the density values of the region-of-interest image P and the restored image Q.

[0099] Further, in the foreign matter inspection device **1** according to the present embodiment, the difference evaluation value obtained from the positive difference δ between the density values is set to be smaller than the difference evaluation value obtained from the negative difference δ between the density values having the same absolute value. Therefore, when the inspection image A is the X-ray absorption image, it is possible to accurately detect the foreign matter having a negative difference between the density values of the region-of-interest image P and the restored image Q.

[0100] Furthermore, in the foreign matter inspection device **1** according to the present embodiment, the difference evaluation value obtained from the negative difference δ between the brightness values is set to be larger than the difference evaluation value obtained from the positive difference δ between the brightness values having the same absolute value. Therefore, when the inspection image A is the X-ray transmission image, it is possible to accurately detect the foreign matter having a negative difference between the brightness values of the region-of-interest image P and the restored image Q.

[0101] Moreover, in the foreign matter inspection device **1** according to the present embodiment, the difference evaluation value obtained from the negative difference δ between the brightness values is set to be smaller than the difference evaluation value obtained from the positive difference δ between the brightness values having the same absolute value. Therefore, when the inspection image A is the X-ray transmission image, it is possible to accurately detect the foreign matter having a positive difference between the brightness values of the region-of-interest image P and the restored image Q.

DESCRIPTION OF REFERENCE NUMERALS AND SIGNS

[0102] **1**: foreign matter inspection device
[0103] **20**: X-ray inspection unit
[0104] **31**: inspection image generation unit
[0105] **32**: foreign matter candidate region specification unit
[0106] **33**: provisional determination unit
[0107] **34**: region-of-interest setting unit
[0108] **35**: mask image generation unit
[0109] **36**: restored image generation unit
[0110] **37**: difference evaluation value calculation unit
[0111] **38**: foreign matter determination unit
[0112] **39**: storage unit
[0113] **45**: display unit
[0114] **46**: operation unit
[0115] **50**: control unit
[0116] **100**: inspection object
[0117] A: inspection image (original image)
[0118] B, B**1**, B**2**: foreign matter candidate region
[0119] M: mask image
[0120] P, P**1**, P**2**: region of interest (region-of-interest image)
[0121] Q: restored image
[0122] W: image of inspection object

What is claimed is:

1. A foreign matter inspection device that detects a foreign matter included in an inspection object based on an image, the foreign matter inspection device comprising:

a region-of-interest setting unit that sets a region of interest including a foreign matter candidate region, which is likely to include an image of the foreign matter in the inspection object, in an inspection image in which the foreign matter candidate region has been specified;

a mask image generation unit that generates a mask image obtained by masking the foreign matter candidate region in an image of the region of interest;

a restored image generation unit that generates a restored image from the mask image based on a restoration algorithm of a model obtained by learning a good product image of the inspection object;

a difference evaluation value calculation unit that calculates a difference evaluation value for evaluating a difference between the image of the region of interest and the restored image; and

a foreign matter determination unit that determines whether or not the foreign matter candidate region includes the image of the foreign matter based on a comparison between the difference evaluation value and a predetermined threshold value.

2. The foreign matter inspection device according to claim **1**,

wherein the inspection image is an X-ray absorption image obtained by converting an amount of X-rays transmitted through the inspection object into a density value proportional to a thickness of the inspection object,

the difference evaluation value is a value based on a difference between density values of the image of the region of interest and the restored image, and

when an absolute value of a positive difference between density values is equal to an absolute value of a negative difference between density values, the difference evaluation value obtained from the positive difference between the density values is different from the difference evaluation value obtained from the negative difference between the density values.

3. The foreign matter inspection device according to claim **1**,

wherein the inspection image is an X-ray transmission image obtained by converting an amount of X-rays transmitted through the inspection object into a brightness value,

the difference evaluation value is a value based on a difference between brightness values of the image of the region of interest and the restored image, and

when an absolute value of a positive difference between brightness values is equal to an absolute value of a negative difference between brightness values, the difference evaluation value obtained from the negative difference between the brightness values is different from the difference evaluation value obtained from the positive difference between the brightness values.

4. The foreign matter inspection device according to claim **1**, further comprising:

a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image.

5. The foreign matter inspection device according to claim **2**, further comprising:

a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image.

6. The foreign matter inspection device according to claim **3**, further comprising:

a foreign matter candidate region specification unit that specifies the foreign matter candidate region in the inspection image.

7. A foreign matter inspection method for detecting a foreign matter included in an inspection object based on an image, the foreign matter inspection method comprising:

a region-of-interest setting step of setting a region of interest including a foreign matter candidate region, which is likely to include an image of the foreign matter in the inspection object, in an inspection image in which the foreign matter candidate region has been specified;

a mask image generation step of generating a mask image obtained by masking the foreign matter candidate region in an image of the region of interest;

a restored image generation step of generating a restored image from the mask image based on a restoration algorithm of a model obtained by learning a good product image of the inspection object;

a difference evaluation value calculation step of calculating a difference evaluation value for evaluating a difference between the image of the region of interest and the restored image; and

a foreign matter determination step of determining whether or not the foreign matter candidate region includes the image of the foreign matter based on a comparison between the difference evaluation value and a predetermined threshold value.

8. The foreign matter inspection method according to claim **7**,

wherein the inspection image is an X-ray absorption image obtained by converting an amount of X-rays transmitted through the inspection object into a density value proportional to a thickness of the inspection object,

the difference evaluation value is a value based on a difference between density values of the image of the region of interest and the restored image, and

when an absolute value of a positive difference between density values is equal to an absolute value of a negative difference between density values, the difference evaluation value obtained from the positive difference between the density values is different from the difference evaluation value obtained from the negative difference between the density values.

9. The foreign matter inspection method according to claim **7**,

wherein the inspection image is an X-ray transmission image obtained by converting an amount of X-rays transmitted through the inspection object into a brightness value,

the difference evaluation value is a value based on a difference between brightness values of the image of the region of interest and the restored image, and

when an absolute value of a positive difference between brightness values is equal to an absolute value of a negative difference between brightness values, the difference evaluation value obtained from the negative difference between the brightness values is different from the difference evaluation value obtained from the positive difference between the brightness values.

10. The foreign matter inspection method according to claim **7**, further comprising:

a foreign matter candidate region specification step of specifying the foreign matter candidate region in the inspection image.

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